Practical – 8

Aim: To develop a mini AI application such as a text classification system, chatbot prototype, or perception demo and understand the ethical considerations, transparency, fairness, and explainability involved in AI systems.

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report

df = pd.read_csv("student_course_reviews_500.csv")
df.head(10)

df.shape
(500,2)

df.info()

df.isna().any()

df.isna().sum()

df_clean =df.dropna()
df_clean.shape

df_clean.isna().sum()

df_clean['sentiment'].unique()

df_clean['sentiment'].value_counts()

 x =df_clean['review_text']
 y =df_clean['sentiment']

x_train,x_test,y_train,y_test =train_test_split(x,y,test_size =0.3,random_state =42,stratify =y)
```

```
vectorizer =CountVectorizer()
x_train_vec = vectorizer.fit_transform(x_train)
x_test_vec =vectorizer.transform(x_test)

model =MultinomialNB()
model.fit(x_train_vec,y_train)

y_pred =model.predict(x_test_vec)

print(classification_report(y_test,y_pred))
new_reviews = [
    "The professor explained every single concept very clearly.",
    "Exams were unreasonably long and grading was completely unfair.",
    "Decent lectures, matched standard course content."
]

new_reviews_vec = vectorizer.transform(new_reviews)
predictions = model.predict(new_reviews_vec)

for review, pred in zip(new_reviews, predictions):
    print(f"Review: '{review}' --> Predicted Sentiment: {pred}")
```

OUTPUT:

	review_text	sentiment
0	Loved how the professor connected theoretical ...	Positive
1	Lectures often went off-topic and missed core ...	Negative
2	Syllabus was updated with latest tools and ind...	Positive
3	Great peer learning environment and active stu...	Positive
4	Neutral opinion overall, it meets basic course...	Neutral
5	Note: Excellent support during office hours fr...	Positive
6	I loved the practical hands-on projects; they ...	Positive
7	Felt like a complete waste of time and tuition...	Negative
8	Very good learning experience overall, worth t...	Positive
9	Terrible communication from the professor rega...	Negative

```
(500, 2)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 2 columns):
 #   Column       Non-Null Count  Dtype 
---  ------       --------------  ----- 
 0   review_text  475 non-null    object
 1   sentiment    485 non-null    object
dtypes: object(2)
memory usage: 7.9+ KB

 0
review_text True
sentiment True

dtype: bool

 0
review_text 25
sentiment 15

dtype: int64

(460, 2)


0
review_text 0
sentiment 0

dtype: int64

array(['Positive', 'Negative', 'Neutral'], dtype=object)
count
sentiment
Positive 184
Negative 177
Neutral 99

dtype: int64
```

```
KNeighborsClassifier
KNeighborsClassifier(n_neighbors=8)
```

	precision	recall	f1-score	support
Negative	0.98	1.00	0.99	53
Neutral	1.00	1.00	1.00	30
Positive	1.00	0.98	0.99	55
accuracy			0.99	138
macro avg	0.99	0.99	0.99	138
weighted avg	0.99	0.99	0.99	138

```
Review: 'The professor explained every single concept very clearly.' --> Predicted Sentiment: Positive
Review: 'Exams were unreasonably long and grading was completely unfair.' --> Predicted Sentiment: Negative
Review: 'Decent lectures, matched standard course content.' --> Predicted Sentiment: Neutral
```

conclusion:

Thus, a mini AI application was successfully developed and tested. The practical helped us understand how AI applications work and make decisions. We also learned the importance of ethics, fairness, transparency, and explainability in developing responsible AI systems.